THE LI SEQUENCE AS A SAMPLED CONTROL SYSTEM: FIR WINDOW STABILITY, OFF-LINE MODES, AND THE ARITHMETIC CLOSURE PROBLEM

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Abstract. We formulate the Li–Keiper sequence of the Riemann zeta function as the sampled output of a discrete-time spectral system. Under the Cayley map

$$w_\rho = 1 - \rho^{-1},$$

a zero on the critical line becomes a unit-circle mode, while a zero to the right becomes an unstable mode w_ρ^{-n}. The symmetric zero orbit has the exact response

$$4 - 4\cosh(na_\rho)\cos(n\theta_\rho).$$

Permanent cancellation of an unstable mode by the remaining zeros is impossible: the exponential rate of the full Li sequence is the reciprocal of the smallest modulus of an interior Cayley pole.

For every fixed moving-window length L, the output

$$Y_L(N) = \sum_{j=0}^{L-1} \lambda_{N+j}$$

is subexponential if and only if RH holds. More strongly, RH is equivalent to eventual nonnegativity of every sliding output $Y_L(N)$. Repeated window averaging is a cascade of FIR filters $H_L(z) = 1 + \cdots + z^{L-1}$. Since all zeros of H_L lie on the unit circle, no such cascade can cancel an off-line mode. We also construct an adjacent-window bank with no blind spot:

$$|H_L(z)|^2 + |H_{L+1}(z)|^2 \geq \tfrac{1}{2}|z|^{2L}.$$

The signal representation lifts exactly to the ordinary von Mangoldt weights. A sliding window collapses the prime–Laguerre kernel to

$$L^{(2)}_{N+L-2}(x) - L^{(2)}_{N-2}(x).$$

Nested linear averages remain FIR filters and therefore detect, but do not bound, every unstable mode. A rational off-line quartet proves that a signed block average may be negative while a pointwise exponential violation occurs inside the same block. For the deterministic blocks

$$I_L = \{L^2, \ldots, L^2 + L - 1\},$$

we prove that the L^2-th root of $1 + \sum_{n \in I_L} |\lambda_n|$ converges exactly to the largest Cayley modulus of a zeta zero. The remaining arithmetic task is a one-sided dissipativity estimate for the complete prime–pole output, or a nonlinear positive-part estimate. We state that estimate exactly. It is not proved here; accordingly this paper does not claim a proof of RH. Its contribution is a rigorous signals-and-systems architecture, exact RH criteria, and a precise identification of the plant-specific inequality still missing for the literal weights $\Lambda(m)$. A lossless prefix compression reduces the absolute block problem from all 2^L sign selectors to only L differences of two Laguerre boundary degrees.

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1. Introduction

The Riemann Hypothesis is usually presented as a geometric statement about the zero set of $\zeta(s)$: every nontrivial zero should lie on $\operatorname{Re} s = 1/2$. The equivalent Li criterion replaces this infinite geometry by the sign of a sampled sequence $(\lambda_n)_{n\geq 1}$. That replacement suggests a third language, familiar from electrical engineering.

Discrete-time signals and control	Zeta/Li object
sample number	Li degree n
complex modal frequency	$w_\rho = 1 - \rho^{-1}$
unit-circle mode	zero with $\operatorname{Re}\rho = 1/2$
unstable mode	zero with $\operatorname{Re}\rho > 1/2$
sampled plant output	λ_n
FIR moving window	$\sum_{j=0}^{L-1} \lambda_{N+j}$
spectral radius test	$\limsup \lvert\lambda_n\rvert^{1/n}$
one-sided dissipativity	eventual positivity of a window output

The fundamental point is the exact identity

$$\left|1 - \frac{1}{\rho}\right|^2 = 1 + \frac{1 - 2\operatorname{Re}\rho}{|\rho|^2}.$$

The critical line maps to the unit circle. A zero to the right maps inside the circle, and the coefficient sequence sees its reciprocal, an unstable mode. In this language RH asks whether the zeta plant is marginally stable: polynomial growth is allowed, but a fixed exponential rate is not.

The formulation separates two tasks. The zero-side task is to prove that an off-line zero creates an unstable output and cannot be hidden forever by modal interference. We complete it in Sections 2 and 6. The arithmetic-side task is to prove, from the literal weights $\Lambda(m)$, that a suitable filtered output is subexponential or one-sided dissipative. We derive its exact kernel in Section 7. That second estimate remains open.

This distinction is essential. Computation of finitely many critical zeros tests the response of known unit-circle modes, but does not control an unknown unstable mode above the verified height. Conversely, an FIR filter can be proved to detect every unstable mode without thereby proving that the actual plant has none. Detection and exclusion are separate steps.

2. Li coefficients and the Cayley plant

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

be the completed Riemann function. The Li coefficients are given by

$$\lambda_n = \sum_\rho \left\{1 - \left(1 - \frac{1}{\rho}\right)^n\right\}, \tag{2.1}$$

with the standard symmetric interpretation of the zero sum. Equivalently,

$$G(z) := z\frac{d}{dz}\log\xi\left(\frac{1}{1-z}\right) = \sum_{n\geq 1}\lambda_n z^n. \tag{2.2}$$

Lemma 2.1 (Cayley modulus). *For $\rho = \beta + i\gamma$ and $w_\rho = 1 - \rho^{-1}$,*

$$|w_\rho|^2 = 1 + \frac{1-2\beta}{|\rho|^2}. \tag{2.3}$$

Thus $|w_\rho| = 1$ exactly on the critical line and $|w_\rho| < 1$ exactly when $\beta > 1/2$.

Proof. Direct expansion gives

$$|w_\rho|^2 = \frac{(\beta-1)^2+\gamma^2}{\beta^2+\gamma^2} = 1 + \frac{1-2\beta}{\beta^2+\gamma^2}.$$

□

3. FIR windows and filtered Li outputs

For a finite real impulse response

$$h = (h_0, \dots, h_{L-1}),$$

define

$$H_h(z) = \sum_{j=0}^{L-1} h_j z^j, \qquad Y_h(N) = \sum_{j=0}^{L-1} h_j \lambda_{N+j}. \tag{3.1}$$

This is a forward-index convention. A mode w^{-n} is multiplied by $H_h(w^{-1})$.

Theorem 3.1 (No-cancellation FIR criterion). *Assume that $H_h(z) \neq 0$ for every $|z| > 1$. Then*

$$\mathrm{RH} \iff \limsup_{N\to\infty} |Y_h(N)|^{1/N} \leq 1. \tag{3.2}$$

Proof. Under RH, $Y_h(N) = O_h(N \log N)$. If RH is false, insert (2.10) into (3.1):

$$Y_h(N) = -\sum_{|w|=r_0} m_w w^{-N} H_h(w^{-1}) + O_h(r_1^{-N}). \tag{3.3}$$

Every multiplier in the dominant sum is nonzero. After equal phases are grouped, the same mean-square argument as in Theorem 6.2 shows that the dominant trigonometric polynomial is nonzero. Thus the left side of (3.2) equals $r_0^{-1} > 1$. □

3.1. Moving averages.

The unnormalised moving-window filter is

$$H_L(z) = 1 + z + \cdots + z^{L-1} = \frac{1-z^L}{1-z}. \tag{3.4}$$

Its zeros are the nontrivial L-th roots of unity, so it has no zero outside the unit circle.

Corollary 3.2 (Sliding-window stability criterion). *For each fixed $L \geq 1$, put*

$$Y_L(N) = \sum_{j=0}^{L-1} \lambda_{N+j}. \tag{3.5}$$

Then

$$\boxed{\mathrm{RH} \iff \limsup_{N\to\infty} |Y_L(N)|^{1/N} \leq 1.} \tag{3.6}$$

This criterion does not require every sample λ_n to have a fixed sign. It asks only that one FIR output have no unstable exponential rate.

3.2. **Cascades and averages of averages.** Suppose a window output is averaged again, with lengths $L_1, \dots, L_k$. The resulting transfer function is

$$H_{\boldsymbol{L}}(z) = \prod_{\nu=1}^{k} H_{L_\nu}(z). \tag{3.7}$$

All its zeros remain on the unit circle.

Corollary 3.3 (Cascade invariance)**.** *Every finite cascade of moving averages satisfies*

$$\mathrm{RH} \iff \limsup_{N\to\infty} |Y_{\boldsymbol{L}}(N)|^{1/N} \le 1. \tag{3.8}$$

No finite number of linear window averages can stabilize or erase an off-line mode.

Remark 3.4. This proves the useful part of averaging windows and then averaging those averages. The cascade remains a valid detector. It does not prove that the detector output is subexponential for the actual prime weights. A filter transfer identity is not a plant estimate.

4. One-sided window dissipativity

Subexponential stability is bilateral. A one-sided sign condition is sometimes more compatible with arithmetic positivity.

Theorem 4.1 (Eventual moving-window positivity)**.** *For every fixed $L \ge 1$, the following are equivalent:*

1. *RH;*
2. *there exists N_0 such that*

$$Y_L(N) = \sum_{j=0}^{L-1} \lambda_{N+j} \ge 0 \qquad (N \ge N_0). \tag{4.1}$$

Proof. Under RH, Li's criterion gives $\lambda_n \ge 0$ for every n, hence (4.1).

If RH is false, formula (3.3) with $H_h = H_L$ has a nonzero dominant real trigonometric polynomial and exponential factor r_0^{-N}. Every frequency in the polynomial is nontrivial: a nontrivial zeta zero has nonzero ordinate, so its Cayley phase is not zero modulo 2π. On the compact orbit closure the dominant polynomial has Haar mean zero. Since it is not identically zero, it takes both signs. Minimality on the orbit closure supplies infinitely many, in fact syndetically many, returns to each strict sign region. Consequently $Y_L(N)$ has negative exponential excursions and cannot satisfy (4.1). □

Remark 4.2. The quantifier for every sufficiently large sliding position is essential. The existence of some positive long windows does not exclude an off-line mode: the same unstable sinusoid also has positive phases.

5. An adjacent-window filter bank with no blind spot

A single moving average may annihilate a unit-circle frequency. That is not dangerous for detecting off-line modes, but it obscures the total oscillatory content. Two adjacent window lengths remove this blind spot.

Proposition 5.1 (Adjacent-bank coercivity)**.** *For every $z \in \mathbb{C}$ and $L \geq 1$,*

$$H_{L+1}(z) - H_L(z) = z^L, \tag{5.1}$$

and

$$\boxed{|H_L(z)|^2 + |H_{L+1}(z)|^2 \geq \frac{1}{2}|z|^{2L}.} \tag{5.2}$$

Proof. Identity (5.1) is immediate. Since

$$|b-a|^2 \leq 2(|a|^2 + |b|^2),$$

take $a = H_L(z)$, $b = H_{L+1}(z)$ and use (5.1). □

Define the two-channel energy

$$\mathcal{E}_L(N) = |Y_L(N)|^2 + |Y_{L+1}(N)|^2. \tag{5.3}$$

Corollary 5.2 (Two-channel energy criterion)**.** *For every fixed $L \geq 1$,*

$$\mathrm{RH} \iff \limsup_{N\to\infty} \mathcal{E}_L(N)^{1/(2N)} \leq 1. \tag{5.4}$$

Proof. Under RH both channels have polynomial growth. If RH is false, Theorem 6.2 and (5.2) show that the dominant off-line modal vector cannot vanish in both channels. Its exponential rate is $r_0^{-1} > 1$. □

Inequality (5.2) is a genuine observability inequality. It solves the phase-blindness problem of a single window bank. It does not solve the arithmetic upper-bound problem for $\mathcal{E}_L(N)$.

6. Global modal analysis

The functional equation and conjugation complete a noncritical zero to

$$\rho, \quad \overline{\rho}, \quad 1-\rho, \quad 1-\overline{\rho}. \tag{2.4}$$

Their Cayley images are

$$w, \quad \overline{w}, \quad \overline{w}^{-1}, \quad w^{-1}. \tag{2.5}$$

Theorem 6.1 (Exact quartet response)**.** *Suppose* $\operatorname{Re}\rho > 1/2$, *and write*

$$w_\rho = e^{-a+i\theta}, \qquad a > 0.$$

The contribution of the orbit (2.4) *to* λ_n *is*

$$Q_n(\rho) = 4 - 4\cosh(na)\cos(n\theta). \tag{2.6}$$

It has exponentially large excursions. On the critical line the orbit coalesces into a conjugate pair whose contribution is

$$2 - 2\cos(n\theta) \in [0,4]. \tag{2.7}$$

Proof. Using (2.5),

$$\begin{aligned} Q_n(\rho) &= 4 - \{w^n + \overline{w}^n + w^{-n} + \overline{w}^{-n}\} \\ &= 4 - 2(e^{-na} + e^{na})\cos(n\theta), \end{aligned}$$

which is (2.6). Returns of $n\theta$ to a fixed neighbourhood of zero give $\cos(n\theta) \geq 1/2$ infinitely often, and then

$$Q_n(\rho) \leq 4 - e^{na}.$$

For irrational rotation these returns are syndetic; for rational rotation they are periodic. If $a = 0$, the formal quartet is twice the actual conjugate pair, giving (2.7). $\square$

Theorem 6.2 (Global exponential rate)**.** *If RH is false, define*

$$r_0 = \min_{\operatorname{Re}\rho > 1/2} \left| 1 - \frac{1}{\rho} \right| < 1. \tag{2.8}$$

Then

$$\limsup_{n\to\infty} |\lambda_n|^{1/n} = r_0^{-1} > 1. \tag{2.9}$$

More precisely, for some $r_1 \in (r_0, 1)$,

$$\lambda_n = -r_0^{-n} P(n) + O(r_1^{-n}), \tag{2.10}$$

where P is a nonzero finite real trigonometric polynomial.

Proof. The substitution $s = (1 - z)^{-1}$ maps $\mathbb{D}$ onto $\operatorname{Re} s > 1/2$. A zero ρ of multiplicity m_ρ produces in (2.2) a simple pole at w_ρ, with

$$\operatorname{Res}_{z=w_\rho} G(z) = m_\rho w_\rho \neq 0. \tag{2.11}$$

The minimum in (2.8) exists because $|w_\rho| \to 1$ as $|\operatorname{Im}\rho| \to \infty$. Therefore the radius of convergence of (2.2) is r_0, and Cauchy–Hadamard proves (2.9).

There are finitely many poles on $|z| = r_0$. Subtract their principal parts and continue to a circle of radius $r_1 > r_0$ before the next pole. Coefficient extraction gives

$$\lambda_n = -\sum_{|w|=r_0} m_w w^{-n} + O(r_1^{-n}),$$

which is (2.10). Grouping equal phases, the Cesaro mean square of $P(n)$ equals the sum of the squares of its nonzero grouped coefficients. Hence P is bounded away from zero on an infinite subsequence. $\square$

Corollary 6.3 (Two-sided syndetic excursions)**.** *If RH is false, there exist $c > 0$, $R > 1$, and two syndetic sets $D_+, D_- \subset \mathbb{N}$ such that, for every sufficiently large index in the respective set,*

$$\lambda_n \geq cR^n \quad (n \in D_+), \qquad \lambda_n \leq -cR^n \quad (n \in D_-). \tag{2.12a}$$

Proof. In (2.10), restrict the dominant real trigonometric polynomial to the compact closure of its phase orbit. Its Haar mean is zero and it is not identically zero, so it has two nonempty strict sign regions. Return times to either open region are syndetic by minimality of the orbit rotation. The smaller-radius remainder is absorbed after increasing the starting index. $\square$

Corollary 6.4 (Marginal-stability criterion)**.**

$$\text{RH} \iff \limsup_{n\to\infty} |\lambda_n|^{1/n} \leq 1. \tag{2.12}$$

Proof. The reverse implication is Theorem 6.2. Under RH, the standard Li asymptotic gives $\lambda_n = O(n \log n)$. $\square$

7. Exact lift to the ordinary prime weights

We now express the FIR output before passing to zeros. Let

$$A_n = 1 - \frac{n}{2}\{\gamma + \log(4\pi)\} + \sum_{\substack{\ell \geq 1 \\ \ell \text{ odd}}} \left\{ \left(1 - \frac{1}{\ell}\right)^n - 1 + \frac{n}{\ell} \right\} \tag{6.1}$$

be the archimedean Li block. For $\varepsilon > 0$, define

$$\begin{aligned} p_n(\varepsilon) &= n \sum_{j=1}^{n} \binom{n-1}{j-1} \frac{(-1)^{j-1}}{j\varepsilon^j}, \\ Q_{n,\varepsilon} &= \sum_{m \geq 2} \frac{\Lambda(m)}{m^{1+\varepsilon}} L_{n-1}^{(1)}(\log m), \\ \lambda_{n,\varepsilon} &= A_n + p_n(\varepsilon) - Q_{n,\varepsilon}. \end{aligned} \tag{6.2}$$

The complete prime–pole cancellation gives

$$\lambda_n = \lim_{\varepsilon \downarrow 0} \lambda_{n,\varepsilon}. \tag{6.3}$$

Lemma 7.1 (Laguerre window collapse). *For $N \geq 2$ and $L \geq 1$,*

$$\sum_{j=0}^{L-1} L_{N+j-1}^{(1)}(x) = L_{N+L-2}^{(2)}(x) - L_{N-2}^{(2)}(x). \tag{6.4}$$

Proof. Use

$$\sum_{k=0}^{M} L_k^{(\alpha)}(x) = L_M^{(\alpha+1)}(x)$$

at $M = N + L - 2$ and $M = N - 2$, and subtract. □

Set

$$\begin{aligned} A_{N,L} &= \sum_{j=0}^{L-1} A_{N+j}, \\ p_{N,L}(\varepsilon) &= \sum_{j=0}^{L-1} p_{N+j}(\varepsilon), \\ K_{N,L}(x) &= L_{N+L-2}^{(2)}(x) - L_{N-2}^{(2)}(x). \end{aligned} \tag{6.5}$$

Theorem 7.2 (Prime–Laguerre FIR identity). *For every fixed $N \geq 2$ and $L \geq 1$,*

$$\boxed{Y_L(N) = \lim_{\varepsilon \downarrow 0} \left[A_{N,L} + p_{N,L}(\varepsilon) - \sum_{m \geq 2} \frac{\Lambda(m)}{m^{1+\varepsilon}} K_{N,L}(\log m) \right]}. \tag{6.6}$$

All primes, prime powers, the pole and the archimedean block remain coupled until after the FIR operation.

Proof. Sum (6.2) over $n = N, \ldots, N + L - 1$, use Lemma 7.1, and then apply (6.3). □

Corollary 7.3 (Exact arithmetic window criterion). *For every fixed $L \geq 1$, RH is equivalent to the existence of N_0 such that, for all $N \geq N_0$,*

$$\lim_{\varepsilon \downarrow 0} \left[A_{N,L} + p_{N,L}(\varepsilon) - \sum_{m \geq 2} \frac{\Lambda(m)}{m^{1+\varepsilon}} K_{N,L}(\log m) \right] \geq 0. \tag{6.7}$$

8. Why repeated linear averaging does not close the estimate

The moving average improves phase variation and retains all unstable modes. These are useful facts, but neither determines the sign of (6.6).

8.1. A rational off-line control.

8.1. **A rational off-line control.** Let

$$w = \frac{4i}{5}, \qquad \rho = \frac{1}{1-w} = \frac{25+20i}{41}. \tag{7.1}$$

Then $\operatorname{Re}\rho > 1/2$ and $|w| = 4/5$. Its quartet contribution is

$$q_n = 4 - 2\operatorname{Re}(w^n + w^{-n}) = \begin{cases} 4 - 2\{(5/4)^n + (4/5)^n\}, & n \equiv 0 \pmod 4, \\ 4 + 2\{(5/4)^n + (4/5)^n\}, & n \equiv 2 \pmod 4, \\ 4, & n \text{ odd}. \end{cases} \tag{7.2}$$

Every four consecutive sufficiently late samples contain a positive exponential violation. Nevertheless, for blocks aligned at $4K+1$,

$$\sum_{n=4K+1}^{4K+4M} q_n \longrightarrow -\infty \qquad (K \to \infty) \tag{7.3}$$

for each fixed $M \geq 1$. Thus a signed mean can have the desired upper sign while hiding a bad sample in every four positions.

Proposition 8.1 (Linear-cascade limitation). *No inference from the sign or boundedness of a signed linear block mean to a pointwise block barrier is valid on the class of symmetric zero orbits. Repeating the average does not repair the inference: it only replaces H_L by a product of moving-average polynomials, and the off-line mode survives by Corollary 3.3.*

The control (7.1) is not claimed to arise from the ordinary Euler product. It falsifies a signal-processing inference, not the desired arithmetic inequality.

8.2. **Energy also needs an upper bound.** Squaring a filter-bank output prevents sign cancellation: $\mathcal{E}_L(N) \geq 0$. But nonnegativity is the wrong orientation. To exclude an unstable mode one needs an upper bound

$$\mathcal{E}_L(N) \leq e^{o(N)}. \tag{7.4}$$

After insertion of (6.6), this square contains separately divergent prime and polar channels. Expanding them before the Abel limit destroys the cancellation that defines λ_n. Parseval gives an exact detector, but not estimate (7.4).

9. Robust nonlinear statistics of the modal growth rate

The raw values λ_n are signed and may differ by exponential factors, so robust estimators should not be applied to them directly. The scale-invariant sample is

$$a_n := \frac{1}{n}\log(1+|\lambda_n|) \geq 0. \tag{8.1}$$

Critical behaviour means $a_n \to 0$. An off-line mode gives $a_n \geq c > 0$ on a syndetic set.

9.1. **Why fixed trimming and the median are blind.** The syndetic excursion set in Corollary 6.3 has positive density, but there is no universal positive lower bound for that density over all possible phase systems. A median discards every exceptional set of density below 1/2. A trimmed mean that removes a fixed fraction $\tau > 0$ of the largest observations can likewise remove an off-line excursion set whose density is smaller than τ.

The correct robust operation is a vanishing trim. Let

$$I_L = \{L^2, L^2+1, \ldots, L^2+L-1\}, \tag{8.2}$$

and write the corresponding samples in increasing order,

$$a_{L,(1)} \leq \cdots \leq a_{L,(L)}.$$

Choose integers $k_L = o(L)$, and define

$$\begin{aligned} \mathrm{TM}_L &= \frac{1}{L-k_L}\sum_{j=1}^{L-k_L} a_{L,(j)}, \\ \mathrm{Q}_L &= a_{L,(L-k_L)}. \end{aligned} \tag{8.3}$$

Theorem 9.1 (Vanishing-trim criterion)**.** *For every sequence $k_L = o(L)$,*

$$\boxed{\mathrm{RH} \iff \mathrm{TM}_L \longrightarrow 0 \iff \mathrm{Q}_L \longrightarrow 0.} \tag{8.4}$$

Proof. Under RH, $\lambda_n = O(n\log n)$, hence

$$\max_{n\in I_L} a_n \longrightarrow 0.$$

Both statistics therefore tend to zero.

If RH is false, Corollary 6.3 supplies $c > 0$, $R > 1$ and a syndetic set D on which $|\lambda_n| \geq cR^n$. If G is a gap bound for D, then every sufficiently large interval of length L contains at least $L/G - O(1)$ members of D. On those members,

$$a_n \geq \frac{1}{2}\log R$$

for all sufficiently large n. Since $k_L = o(L)$, deleting the k_L largest samples leaves at least $L/G - o(L)$ such observations. Consequently

$$\liminf_L \mathrm{Q}_L \geq \frac{1}{2}\log R, \qquad \liminf_L \mathrm{TM}_L \geq \frac{1}{2G}\log R > 0.$$

$\square$

9.2. **A geometric-mean criterion.** Define the geometric mean of the per-sample growth factors by

$$\mathrm{GM}_L = \left\{ \prod_{n\in I_L} (1+|\lambda_n|)^{1/n} \right\}^{1/L}. \tag{8.5}$$

Its logarithm is $L^{-1}\sum_{n\in I_L} a_n$.

Corollary 9.2 (Geometric growth criterion)**.**

$$\boxed{\mathrm{RH} \iff \mathrm{GM}_L \to 1.} \tag{8.6}$$

Proof. Under RH the maximum of the a_n on I_L tends to zero. Under non-RH, the syndetic set used in Theorem 9.1 contributes a positive lower density of samples with $a_n \geq \frac{1}{2}\log R$. Thus $\liminf \log \mathrm{GM}_L > 0$. □

9.3. **Exact block spectral radius.** The robust statistic above has a sharper unlogged form. Put

$$S_L = \sum_{n=L^2}^{L^2+L-1} |\lambda_n|, \qquad \mathcal{R}_\zeta = \max\left\{1, \max_\rho \left|1-\frac{1}{\rho}\right|\right\}. \tag{8.7}$$

If an off-line zero exists, the maximum is attained because the Cayley moduli tend to one with the zero height.

Theorem 9.3 (Block spectral-radius identity)**.**

$$\boxed{\lim_{L\to\infty} (1+S_L)^{1/L^2} = \mathcal{R}_\zeta.} \tag{8.8}$$

In particular,

$$\boxed{\mathrm{RH} \quad\iff\quad \log(1+S_L) = o(L^2).} \tag{8.9}$$

Proof. Under RH, the estimate in the proof of Corollary 6.4 can be obtained directly by splitting the critical zeros at height n:

$$\lambda_n \leq 4N(n) + n^2 \sum_{\gamma>n} \gamma^{-2} \ll n\log(n+2).$$

Thus $S_L \ll L^3 \log(L+2)$, which gives the value one in (8.8).

Suppose RH is false. Use the notation of Theorem 6.2, so that

$$\lambda_n = -R^n P(n) + O(R_1^n), \qquad R = r_0^{-1} > R_1,$$

where, after equal phases have been grouped,

$$P(n) = \sum_{j=1}^{d} M_j u_j^n, \qquad |u_j| = 1, \quad u_j \neq u_k.$$

Uniformly in the initial index N, finite geometric summation gives

$$\sum_{n=N}^{N+L-1} |P(n)|^2 = L\sum_j M_j^2 + O_P(1). \tag{8.10}$$

Hence every sufficiently long block contains an n for which $|P(n)| \geq c_P > 0$. Taking $N = L^2$, the smaller exponential in the remainder is absorbed and

$$cR^{L^2} \leq S_L \leq CLR^{L^2+L}. \tag{8.11}$$

It follows that $L^{-2} \log S_L \to \log R$. Functional symmetry pairs the innermost Cayley pole with the largest reciprocal modulus, so $R = \mathcal{R}_\zeta$. This proves (8.8)–(8.9). □

Remark 9.4. Winsorisation with a fixed clipping fraction has the same blindness as fixed trimming. A harmonic mean is dominated by samples near zero and can miss an unstable oscillatory mode. Poisson, negative-binomial and Bayesian count models require stochastic assumptions not supplied by the zeta problem. The deterministic compact-group return theorem, rather than an independence model, is the relevant substitute.

10. The nonlinear closure problem

There are two exact versions of the missing estimate.

10.1. **One-sided dissipativity for a fixed FIR output.**

The cleanest signals-and-systems target is

$$Y_L(N) \geq 0 \quad (N \geq N_0). \tag{9.1}$$

It is weaker than coefficientwise Li positivity because individual λ_n may be negative and be compensated inside the moving window. It is nevertheless equivalent to RH because the FIR transfer has no zero off the unit circle.

10.2. **Positive-part supply on growing blocks.**

For a growing interval $I = [N, N + M - 1]$, define

$$\mathcal{S}(I) = \sum_{n \in I} (\lambda_n - e^{\sqrt{n}})_+. \tag{9.2}$$

The pointwise upper barrier holds on I exactly when $\mathcal{S}(I) = 0$. Its arithmetic representation is

$$\mathcal{S}(I) = \lim_{\varepsilon \downarrow 0} \sum_{n \in I} \left(A_n + p_n(\varepsilon) - \sum_{m \geq 2} \frac{\Lambda(m)}{m^{1+\varepsilon}} L^{(1)}_{n-1}(\log m) - e^{\sqrt{n}} \right)_+. \tag{9.3}$$

The positive part is taken only after pole, Gamma, primes and all prime powers have been combined.

10.3. **Absolute-block selector.**

The block norm in Theorem 9.3 also has a literal-prime form which keeps the two divergent channels coupled. Define

$$d\nu_\varepsilon(u) = e^{-\varepsilon u}\, du - \sum_{m \geq 2} \frac{\Lambda(m)}{m^{1+\varepsilon}} \delta_{\log m}(du), \tag{9.4}$$

and, for $\sigma \in [-1, 1]^L$,

$$P_{\sigma,L}(u) = \sum_{j=0}^{L-1} \sigma_j L^{(1)}_{L^2+j-1}(u). \tag{9.5}$$

Since

$$p_n(\varepsilon) = \int_0^\infty e^{-\varepsilon u} L^{(1)}_{n-1}(u)\, du,$$

finite-dimensional ℓ^1–ℓ^∞ duality gives

$$\boxed{S_L = \lim_{\varepsilon\downarrow 0} \sup_{\sigma\in[-1,1]^L} \left\{ \sum_{j=0}^{L-1} \sigma_j A_{L^2+j} + \int_0^\infty P_{\sigma,L}(u)\, d\nu_\varepsilon(u) \right\}}. \tag{9.6}$$

The interchange is legitimate because the block is finite: the difference between the two suprema is at most $\sum_{j<L} |\lambda_{L^2+j,\varepsilon} - \lambda_{L^2+j}|$. Consequently, the direct ordinary-prime closure corresponding to (8.9) is

$$\log\left[1 + \lim_{\varepsilon\downarrow 0} \sup_{\sigma\in[-1,1]^L} \left\{ \sum_{j<L} \sigma_j A_{L^2+j} + \int P_{\sigma,L}\, d\nu_\varepsilon \right\}\right] = o(L^2). \tag{9.7}$$

By Theorem 9.3, (9.7) is exactly equivalent to RH. It is not inferred by placing separate moduli on the continuum and prime parts of (9.4).

10.4. **Lossless prefix compression.** Put $N = L^2$ and

$$T_{L,j} = \sum_{r=0}^{j} \lambda_{N+r}, \qquad M_L = \max_{0\le j<L} |T_{L,j}|. \tag{9.8}$$

Proposition 10.1 (Prefix compression)**.** *For every $L \ge 1$,*

$$\boxed{M_L \le S_L \le (2L-1)M_L.} \tag{9.9}$$

Consequently

$$\boxed{\mathrm{RH} \iff \log(1+M_L) = o(L^2).} \tag{9.10}$$

Proof. The first inequality is the triangle inequality. Conversely, if $x_r = \lambda_{N+r}$, then $x_0 = T_{L,0}$ and $x_r = T_{L,r} - T_{L,r-1}$ for $r \ge 1$. Hence

$$\sum_{r<L} |x_r| \le |T_{L,0}| + \sum_{r=1}^{L-1} (|T_{L,r}| + |T_{L,r-1}|) \le (2L-1)M_L.$$

Since $\log(2L-1) = o(L^2)$, (9.10) follows from Theorem 9.3. □

This elementary compression has a useful exact arithmetic consequence. Define

$$\Delta(u) = \gamma - u + \sum_{m\le e^u} \frac{\Lambda(m)}{m}, \qquad H_{N,j}(u) = L^{(3)}_{N+j-2}(u) - L^{(3)}_{N-3}(u). \tag{9.11}$$

Abel summation of the coupled measure (9.4), followed by

$$\sum_{r=0}^{j} L^{(1)}_{N+r-1} = L^{(2)}_{N+j-1} - L^{(2)}_{N-2}, \tag{9.12}$$

gives

$$\begin{aligned} T_{L,j} = {} & \sum_{r=0}^{j} A_{N+r} + \gamma \left\{ \binom{N+j+1}{2} - \binom{N}{2} \right\} \\ & - \int_0^\infty H_{N,j}(u)\Delta(u)\, du. \end{aligned} \tag{9.13}$$

The two explicit terms are $\exp\{o(N)\}$, uniformly for $j < L = \sqrt{N}$. Therefore the remaining ordinary-prime estimate is precisely

$$\boxed{\log\left(1 + \max_{0\le j<L}\left|\int_0^\infty \big(L^{(3)}_{L^2+j-2} - L^{(3)}_{L^2-3}\big)(u)\left(\gamma - u + \sum_{m\le e^u}\frac{\Lambda(m)}{m}\right)du\right|\right) = o(L^2).} \tag{9.14}$$

By (9.10)–(9.13), (9.14) is equivalent to RH. Its advantage over (9.7) is combinatorial rather than logical: only L two-boundary kernels remain. No estimate for (9.14) is asserted here.

Open closure problem 10.2 (Arithmetic dissipativity)**.** Prove one of the following directly for the ordinary von Mangoldt weights:

(1) for one fixed L, inequality (9.1) holds for all sufficiently large N;
(2) there exist consecutive intervals I_k, with $|I_k| \to \infty$, such that $\mathcal{S}(I_k) = 0$;
(3) for one fixed L, the adjacent-bank energy satisfies $\mathcal{E}_L(N) \le e^{o(N)}$.

Each assertion proves RH.

Indeed, item 1 closes by Theorem 4.1, and item 3 by Corollary 5.2. For item 2, if RH were false then Corollary 6.3 would provide a syndetic set D_+ on which $\lambda_n \ge cR^n$. Eventually $cR^n > e^{\sqrt{n}}$, and every sufficiently long interval would meet D_+; hence its supply $\mathcal{S}(I)$ would be strictly positive. This contradicts the existence of zero-supply intervals of unbounded length.

The three targets encode unstable-mode exclusion with different nonlinearities: a half-space, a positive-part supply, and an energy upper bound. The first is closest to classical dissipativity; the second is the weakest pointwise block formulation; the third is phase blind.

11. Status table

The results separate observability from stability.

Statement	Status	Meaning
Off-line quartet has exponential envelope	proved	unstable mode exists
Other zeros cannot cancel it forever	proved	global observability
Moving averages preserve it	proved	FIR detector has no off-unit zero
Adjacent windows have no common blind spot	proved	quantitative observability
Vanishing-trim and geometric-rate criteria	proved	robust nonlinear detection
Exact absolute-block spectral radius	proved	block rate equals largest Cayley modulus
Prime–Laguerre window identity	proved	exact arithmetic realization
Signed average implies a good pointwise block	false	rational quartet witness
Ordinary-prime FIR output is subexponential	open	stability estimate
Ordinary-prime FIR output is eventually nonnegative	open	one-sided dissipativity

The fact that all computed zeros are critical validates the marginally stable response over the verified spectral range. It does not prove the last two rows. Similarly, the exact decomposition into primes, pole and Gamma factor identifies every component of the plant, but a decomposition does not provide the signed upper estimate needed to establish stability.

12. Reproducibility

The companion script

`03-research/phase-105-a1-geometry-and-deep-limit/tools/fir_window_bank_check.py`

uses exact rational arithmetic for the Laguerre degree-sum identity and checks the FIR identities, the adjacent-bank inequality, survival of an off-unit mode through a cascade, and the rational signed-average witness. It is run from the repository root with

```
python3 03-research/phase-105-a1-geometry-and-deep-limit/tools/fir_window_bank_check.py.
```

These computations audit algebraic identities; the analytic proofs are the arguments in the preceding sections.

13. Conclusion

The Cayley transform makes the signals-and-systems content of the Li criterion exact. Critical zeros are unit-circle modes. Off-line zeros are unstable modes. The Li coefficients are the sampled output. Moving windows are FIR filters. Repeated averaging is a filter cascade. A pair of adjacent windows is an observable filter bank with the quantitative lower bound (5.2).

For every fixed L,

$$\mathrm{RH} \iff \limsup_{N} \left| \sum_{j=0}^{L-1} \lambda_{N+j} \right|^{1/N} \le 1,$$

and also

$$\mathrm{RH} \iff \sum_{j=0}^{L-1} \lambda_{N+j} \ge 0 \quad \text{for every sufficiently large } N.$$

The corresponding prime-side output uses one difference of two Laguerre polynomials of parameter 2.

The remaining issue is not whether the filter detects an off-line zero; it does. It is whether the literal arithmetic plant satisfies a subexponential upper bound or a one-sided supply inequality. Repeated linear averaging alone cannot prove that bound, and a quadratic detector still requires an upper estimate after exact prime–pole cancellation. Problem 10.2 records the closure in three control-theoretic forms. Solving any one of them would complete a proof of RH.

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